THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Which of these is the best reason for taking repeated measurements? 1

(a) to waste time **(b)** to make the object longer
(c) to reduce random errors **(d)** to change units

2. A bottle cap made of plastic is replaced by a metal cap. Which problem might arise? 1

(a) it floats **(b)** it becomes transparent
(c) it may rust or conduct heat **(d)** it dissolves

3. Which of the following devices uses an insulator to protect the user? 1

(a) electric iron handle **(b)** kettle body
(c) iron rod **(d)** frying pan base

4. Which of the following changes is reversible? 1

(a) melting of ice **(b)** rusting **(c)** cooking of rice **(d)** burning of wood

5. Which of the following correctly orders the steps of a municipal water treatment? 1

(a) sedimentation, filtration, disinfection **(b)** boiling, magnet
(c) disinfection, filtration **(d)** filtration, sedimentation

6. Which characteristic makes 'excretion' essential for life? 1

(a) it makes food **(b)** it helps in growth
(c) accumulated wastes are toxic **(d)** it makes light

7. A village uses a solar pump instead of a diesel pump. The main benefit is: 1

(a) more fuel **(b)** no fuel cost and no smoke
(c) more smoke **(d)** more noise

8. How would the length of day change if the Earth rotated twice as fast? 1

(a) halve **(b)** double **(c)** stay the same **(d)** vanish

9. Which of these is the best example of a testable question? 1

(a) Which sport is exciting?
(b) Does temperature affect how fast sugar dissolves?
(c) Is blue a better colour than red?
(d) Is a mango tastier than a pear?

10. Which group of features suits an animal living in the polar region? 1

(a) feathers, hollow bones, beak **(b)** gills, fins, scales
(c) thick fur, fat layer, white colour **(d)** thin skin, long legs, hump

11. Which pair correctly matches the deficiency and disease? 1

(a) iodine, anaemia **(b)** vitamin A, night blindness
(c) iron, scurvy **(d)** vitamin C, rickets

12. A compass kept on a table shows north in a wrong direction. The best explanation is: 1

(a) the compass is in sunlight **(b)** the table is made of wood
(c) the compass needle is too light **(d)** a magnet or iron object nearby

13. A thread of length 188 cm goes exactly 4 times round the edge of a circular plate. The circumference of the plate is: 1

(a) 752 cm **(b)** 47 cm **(c)** 94 cm **(d)** 188 cm

14. Which of these shows that solubility depends on the substance and not only on appearance? 1

(a) sugar and chalk powder both look white but behave differently in water
(b) both are white
(c) both are powders
(d) both are solid

15. A student records water temperature in the evening: 20 °C, 20 °C, 20 °C after placing in a room at 20 °C. This shows: 1

(a) the water is cooling
(b) the water is boiling
(c) the water is at the same temperature as the room
(d) the water is being heated

16. If the surface area of a liquid is doubled, the rate of evaporation will: 1

(a) increase **(b)** stay the same **(c)** stop **(d)** decrease

17. A mixture of two liquids that do not mix is separated by: 1

(a) winnowing **(b)** filtration through paper
(c) decantation after the layers form **(d)** sieving

18. Which of these cannot be taken as proof of life when used alone? 1

(a) respiration **(b)** growth from inside
(c) reproduction **(d)** movement

19. A hill slope is turned into steps and trees planted. The main effect is: 1

(a) higher hill **(b)** more soil lost
(c) no rain **(d)** less soil washed away

20. On a full-moon night the Moon rises roughly at: 1

(a) sunrise **(b)** noon **(c)** sunset **(d)** midnight

21. C.V. Raman's work on the scattering of light began with his curiosity about why the sea looks blue. This shows that: 1

(a) curiosity about everyday things can lead to great discoveries
(b) experiments are unnecessary in science
(c) only costly equipment can lead to discoveries
(d) great discoveries need no observation

22. The roots of a banyan hanging from branches are called: 1

(a) prop roots **(b)** fibrous roots **(c)** tap roots **(d)** tuberous roots

23. Which food pair shows foods rich in fat and protein respectively? 1

(a) salt and sugar **(b)** rice and butter **(c)** butter and dal **(d)** apple and orange

24. A pin stuck to a magnet can also pick up another pin because: 1

(a) the pin is sticky **(b)** pins have glue
(c) air presses them **(d)** the pin acts as a temporary magnet

25. Which statement about a ceiling fan switched on is correct? 1

(a) the blades move randomly **(b)** each blade moves linearly
(c) the fan is at rest **(d)** each blade moves in a circular path

26. After removing iron filings, adding water to the mixture and filtering separates: 1

(a) sand on the filter paper **(b)** salt on the filter paper
(c) both on the filter **(d)** nothing

27. Which temperature is not possible on a clinical thermometer scale? 1

(a) 20 °C **(b)** 40 °C **(c)** 36 °C **(d)** 42 °C

28. A tap leaks 1 drop per second; if 20 drops make 1 mL, the leak in one hour is about: 1

(a) 18 mL **(b)** 180 mL **(c)** 3600 mL **(d)** 60 mL

29. A mixture of sand and iron in 1:1 by mass weighing 40 g. After using a magnet, the iron obtained should be: 1

(a) 20 g **(b)** 0 g **(c)** 40 g **(d)** 10 g

30. A calf grows into a cow. Its growth involves: 1

(a) melting **(b)** growth from within using food
(c) absorption of sand **(d)** addition of stones

31. A farmer notes that crop yield fell year after year in the same field with the same crop. Probable reason: 1

(a) nutrient loss in soil **(b)** more worms
(c) more sunshine **(d)** more rain

32. A satellite completes 16 orbits in one day (24 hours). The time taken for one orbit is: 1

(a) 90 minutes **(b)** 24 hours **(c)** 60 minutes **(d)** 120 minutes

33. In a graph of plant height against days, a steady upward line shows: 1

(a) the plant is not changing **(b)** the plant is shrinking
(c) the soil is dry **(d)** the plant grows with time

34. The most likely reason that a cactus has thick, fleshy stem is: 1

(a) to hold soil **(b)** to make food in the dark
(c) to attract birds **(d)** to store water

35. A mother wishes to improve her child's diet cheaply. Best advice: 1

(a) give only sweets
(b) give only milk
(c) include pulses, seasonal vegetables and local fruits
(d) buy imported snacks

36. Which of these is the correct way to hold a compass while taking a reading? 1

(a) flat on the palm, away from iron objects **(b)** close to a speaker
(c) tilted near a steel desk **(d)** in a metal box

37. Parallax error is least when the scale is: 1

(a) tilted
(b) viewed from the side
(c) placed far away
(d) placed touching the object and viewed straight

38. Which of these should not be done with glass items? 1

(a) washing with water **(b)** drying
(c) keeping on a shelf **(d)** heating suddenly

39. The difference between heat and temperature, as at this level, is that temperature tells: 1

(a) how hot something is **(b)** how much it weighs
(c) its colour **(d)** its volume

40. A closed bottle with water and air is heated. What happens to the vapour? 1

(a) it freezes **(b)** it increases **(c)** it turns to ice **(d)** it vanishes

41. **Assertion (A):** A mixture of sand and water can be separated by filtration. 1
Reason (R): Salt can be obtained from sea water by evaporation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** Plants take in air through tiny pores called stomata. 1
Reason (R): Fish breathe with the help of gills.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** Coal is a non-renewable resource. 1
Reason (R): Nitrogen is the most abundant gas in air.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The Moon does not produce its own light. 1
Reason (R): We see the Moon because it reflects sunlight.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** Scientists repeat an experiment several times before accepting the result. 1
Reason (R): The result of one trial is always enough to be sure.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The leaves of a cactus are reduced to spines to save water in a desert. 1
Reason (R): Cactus plants have large, broad leaves.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** Iodine solution turns blue-black when it is added to a food containing starch. 1
Reason (R): Iodine solution turns blue-black with proteins also.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** Like poles of two magnets attract each other. 1
Reason (R): A north pole and a south pole attract each other.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The motion of a swing is periodic. 1
Reason (R): A swing repeats its motion after equal intervals of time.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** Sand dissolves in water. 1
Reason (R): Salt dissolves in water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**